FPGA Perceptron for Network Intrusion Detection

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Abstract

This document details the implementation of a hardware-accelerated Single Layer Perceptron on a Nexys A7 FPGA, designed to act as a high-speed Network Intrusion Detection System (IDS). The system processes feature vectors extracted from network traffic, applies pre-trained weights, and classifies packets as either normal traffic or anomalies/intrusions.

1 Algorithm and Model Description

The implementation is based on the classical single-layer perceptron model, utilized for the binary classification of linearly separable data within network traffic.

1.1 Node Operation

The perceptron receives input data representing network packet features and computes a weighted sum, to which a bias term is added. The interpretation space is divided into two linear regions: one yielding an output of 1 (anomaly/intrusion) and the other yielding an output of 0 (normal traffic).

The calculation formula for the weighted sum (net) is:

$$net = w \cdot x + b$$

In the software implementation, the bias (b) is treated as an additional input with a constant value of 1.0 appended to the input vector.

1.2 Activation Function

For the classification decision, a step function (Heaviside) is used. If the weighted sum is greater than or equal to 0, the output is 1; otherwise, it is 0. The activation function is defined as follows:

$$Output = \begin{cases} 1, & \text{if } net \geq 0 \\ 0, & \text{if } net < 0 \end{cases}$$

1.3 Learning Algorithm (Delta Rule)

The model's training is performed iteratively by adjusting the weights using the perceptron learning rule. The error is calculated as the absolute difference between the expected value and the actual output ($|expected - output|$). Weight updates follow the formula below, where the learning rate (α) is set to 0.1:

$$w_{new} = w_{old} + \alpha \cdot (d - y) \cdot x_i$$

2 Hardware Arithmetic Implementation

For the hardware component described in the VHDL files, calculations are performed using 32-bit floating-point arithmetic (addition, subtraction, multiplication). The computations strictly adhere to the IEEE 754 single-precision floating-point standard.

The calculation module implements a pipeline that processes the following stages:

- **Mantissa Alignment:** Comparing the exponents and shifting the mantissa of the smaller number.
- **Effective Calculation:** Adding or subtracting the aligned mantissas.
- **Normalization and Rounding:** Adjusting the result using a Leading Zero Detector (LZC) to format the output according to the standard IEEE 754 representation.

3 IDS Integration and Data Formatting

To validate the system and enable hardware-software interaction, a complete machine learning pipeline was developed:

- **Data Generation:** Involves converting the inputs and the bias into a 32-bit hexadecimal representation (Big Endian) compatible with IEEE 754.
- **Trained Weights:** After training, weights are saved both as readable decimal values (FLOAT_WEIGHTS) and as hexadecimal representations (HEX_IEEE754_WEIGHTS) used for loading into the hardware memory.
- **Real-Time Prediction:** Real network features extracted from PCAP files (using Wireshark and Scapy) are sent to the FPGA via a dedicated UART/USB interface for high-speed, real-time threat predictions.